

ANTHONY MATAR

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EDUCATION

University of Virginia, School of Arts and Sciences;
B.A., Computer Science with Distinction; 3.91 GPA

*Charlottesville, VA
May 2025*

- Relevant Coursework: Machine Learning, Natural Language Processing, Artificial Intelligence, Mathematical Statistics, Mathematics for Data Science, Database Systems, Applied Linear Algebra, Probability, Advanced Software Development

RESEARCH EXPERIENCE

National Institute of Mental Health (NIMH), NIH

Bethesda, MD

IRTA Postbac Research Fellow • Machine Learning Team • Advisor: Dr. Francisco Pereira

Aug 2025-Present

- Develop and evaluate LLM-based tools for psychological assessment, clinical decision support, and theoretical understanding, in collaboration with senior postdoctoral researchers
- Concurrently contribute as an ML/data scientist to clinical surgical outcome prediction research, supporting model development across multi-institutional NSQIP cohorts

Johns Hopkins Medical School, Division of Plastic Surgery

Baltimore, MD

Collaborating ML Researcher • PIs: Dr. Adriana Panayi, Dr. Salih Colakoglu

Dec 2025-Present

- Lead end-to-end ML development for two ongoing manuscripts on breast surgery and tongue cancer complications, spanning preprocessing through deployment of patient-facing risk calculators
- Performed external validation of ORN prediction models in an independent 31-patient cohort

Chang Gung Memorial Hospital, Plastic and Reconstructive Microsurgery Center

Linkou, Taiwan

Collaborating ML Researcher • PI: Dr. Huang-Kai Kao

Jul 2024-Oct 2025

- Built an end-to-end ML workflow to predict osteoradionecrosis after oral tumor excision and fibula free flap reconstruction
- Led preprocessing, univariate analyses, model training/validation, and feature-importance analyses to predict osteoradionecrosis risk in a 274-patient cohort; work contributed to a published peer-reviewed manuscript

Harvard Medical School and Brigham and Women's Hospital, Division of Plastic Surgery

Boston, MA

Collaborating Research Data Scientist • PI: Dr. Adriana Panayi

Oct 2022-Jul 2023

- Cleaned and harmonized large-scale datasets (NSQIP>7.5M records; NHIS>60K records) across hundreds of variables in R
- Summarized pre-, intra-, and postoperative factors plus socioeconomic indicators to evaluate sociodemographic disparities across racial groups; work contributed to a published peer-reviewed manuscript
- Developed data visualizations for various projects with ggplot2

CURRENT PROJECTS

Artificial Intelligence–Based Risk Prediction Models for Complications After Tongue Cancer Surgery

- Developed and validated a suite of ML models (DNN, LightGBM, XGBoost, SVM, logistic regression, stacked ensemble) to predict six postoperative complications after glossectomy for tongue cancer, using 8,266 patients from a 13M+ NSQIP cohort
- Sole technical contributor across the full pipeline: preprocessing, statistical analysis, modeling, validation, interpretability, and deployment
- Built and deployed PRO-Tongue (pro-tongue.streamlit.app) for patient-level risk estimation
- Manuscript pending publication

Risk Prediction for Major Complications After Disease-Directed Breast Surgery: Development and Validation of Interpretable Machine Learning Models

- Developed ML-based risk stratification models for major medical/surgical complications, mortality, unplanned reoperation, and venous thrombosis after disease-directed, non-cosmetic, breast surgery
- Curated and cleaned > 13M entries to define an analytic cohort of 737,730 NSQIP patients (2008–2024) and derive/validate clinically usable risk groups
- Designed the modeling strategy (including logistic regression, gradient-boosted trees, neural networks, and stacked ensembles) and defined a final risk-binning scheme
- Built and deployed PRO-Breast, a web app for patient-level risk estimation: (<https://pro-breast.streamlit.app/>)
- Manuscript under peer review

Development and External Validation of Machine Learning Models for Osteoradionecrosis Risk Prediction

- Extended and refined ML models to predict osteoradionecrosis following fibula free flap reconstruction in oral cancer patients.
- Combined study results from an initial Chang Gung Memorial Hospital cohort (274 patients) with an external validation cohort from Johns Hopkins Hospital (31 patients).
- Evaluated and compared six model families across gradient-boosted trees, KNN, SVMs, neural networks, stacked ensembles, and logistic regression; updated risk-stratification framework to four tiers (very low / low / moderate / high)
- Ongoing manuscript preparation.

PUBLISHED RESEARCH

Matar DY, Mackert GA, **Matar AY**, Chen AC, Panayi AC, Knoedler L, Knoedler S, Yang R, Mady LJ, Kao HK. Artificial intelligence outperforms a nomogram for osteoradionecrosis prognostication following fibula free flap reconstruction in oral cancer patients. J Stomatol Oral Maxillofac Surg. 2025 Oct 1;127(1):102584. doi: 10.1016/j.jormas.2025.102584

Matar DY, Knoedler S, **Matar AY**, Friedrich S, Kiwanuka H, Hamaguchi R, Hamwi CM, Hundeshagen G, Haug V, Kneser U, Ray K, Orgill DP, Panayi AC. Surgical Outcomes and Sociodemographic Disparities Across All Races: An ACS-NSQIP and NHIS Multi-Institutional Analysis of Over 7.5 Million Patients. Ann Surg Open. 2024 Jul 16;5(3):e467. doi: 10.1097/AS9.0000000000000467.

WORK EXPERIENCE

Software Engineer Intern, Iridium Business Solutions

Cape Town, South Africa | Summer 2023

- Worked with team of 4 interns to design and deploy an HR analytics dashboard to an existing web application
- Utilized AWS Lambda & SAM to initialize 6 months of data, calculating metrics stored in DynamoDB with ReactJS
- Constructed ReactJS trigger functions to perform scheduled tasks & real-time metric updates on dashboard

TECHNICAL SKILLS

Programming Languages:

Python, R, SQL, Bash/Shell Scripting

ML & Deep Learning:

Scikit-Learn, PyTorch, Optuna, Keras, Tensorflow, Pandas, NumPy, Matplotlib

Developer Tools:

GitHub, Streamlit, Tableau, Bootstrap, AWS, Heroku, Firebase

ML Techniques & Methodologies:

Supervised learning, ensembles & stacking, deep learning, model interpretability/explainability, hyperparameter optimization